

Santosh Chapagain

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SUMMARY

Data Scientist with experience in constructing end-to-end machine learning pipelines using Python, Pandas, Scikit-Learn, TensorFlow and LangChain. Developed different ML models and RAG-based systems, achieving measurable performance enhancements. Focuses on deploying AI models into real-world applications and optimizing their performance.

SKILLS

Programming & Data Manipulation: Python, Pandas, NumPy

Machine Learning & Modeling: Scikit-Learn, Linear Regression, Logistic Regression, Decision Tree, Random Forest, Lasso, Gradient Boosting, XG Boost, Ridge, SVM, Feature Engineering, Cross Validation

Deep Learning: TensorFlow, CNN, ANN, RNN, LSTM, BiLSTM, Transformer

Generative AI & NLP: NLP, LangChain, RAG, LLMs

Data Management & Visualization: SQL, Matplotlib, Seaborn, Hypothesis Testing

Model Deployment: FastAPI

WORK EXPERIENCE

Tagoraa IT Company — Founder

Dec 2025 – Present

- Co-founded an IT company, contributing to software development and technical architecture

Dreams IT Club Nepal — AI Workshop Instructor

Dec 2025

- Trained 50+ students on AI topics, introducing participants to Large Language Models (LLMs) and practical AI applications.

PROJECTS

YouTube AI Chatbot [\[GitHub\]](#)

- Built a RAG-based chatbot using LLaMA 3.2 and Ollama that answers natural language questions over YouTube transcripts; implemented vector caching to reduce retrieval latency, multi-turn conversational memory, and support for multiple languages — deployable as a local, privacy-first assistant.

Duplicate Question Pair Detection [\[GitHub\]](#)

- Developed and compared 4 distinct NLP models — Fine-tuned Transformer, BiLSTM, Sentence Transformer (semantic similarity), and BoW + Random Forest — to classify duplicate question pairs from the Quora dataset.

Credit Card Fraud Detection [\[GitHub\]](#)

- Developed a full ML pipeline to detect fraudulent transactions from imbalanced datasets. Applied SMOTE oversampling, Standard Scalar normalization, and compared Logistic Regression, Decision Tree and Random Forest models. Final Random Forest model achieved 99.99% accuracy, 100% Recall and 99.98% Precision -deployed to Hugging Face Hub.

EDUCATION

Far-Western University | Dreams College Nepal

Jan 2024 – Present

Bachelor in Computer Science and Information Technology

Gurukul College Nepal

Jan 2022 – Jan 2024

XII (NEB) • GPA: 3.48 • A Grade

CERTIFICATIONS

- Complete Machine Learning , Deep Learning, NLP – CampusX
- Job Ready Data Science Program – CodeWithHarry
- OCI Data Science Race Compition-2025